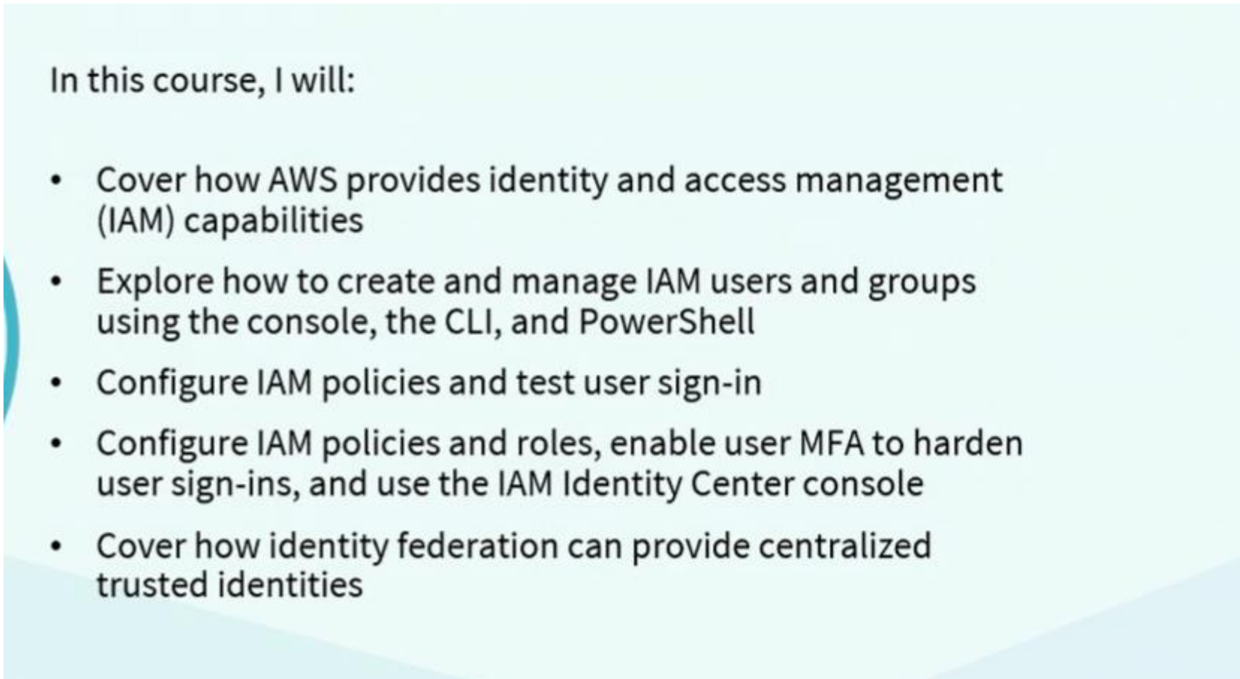
2.1 Identity and Access Management

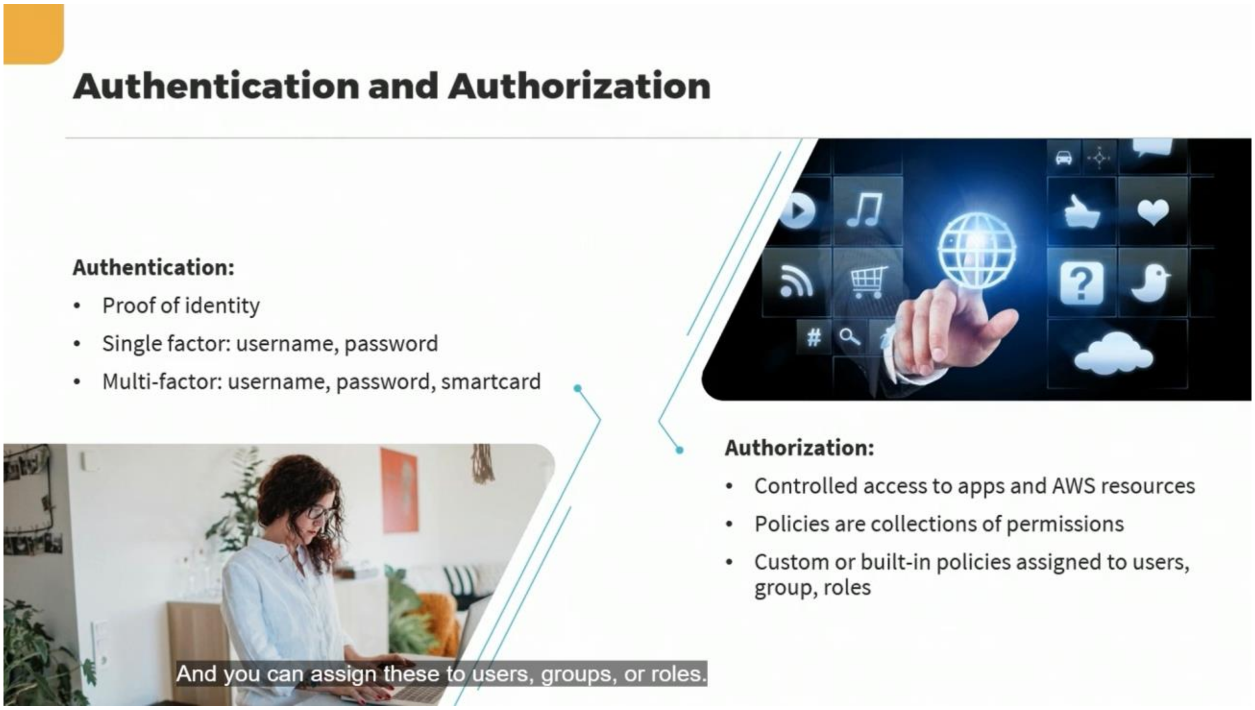


AWS Identity and Access Management (IAM)

Management of:

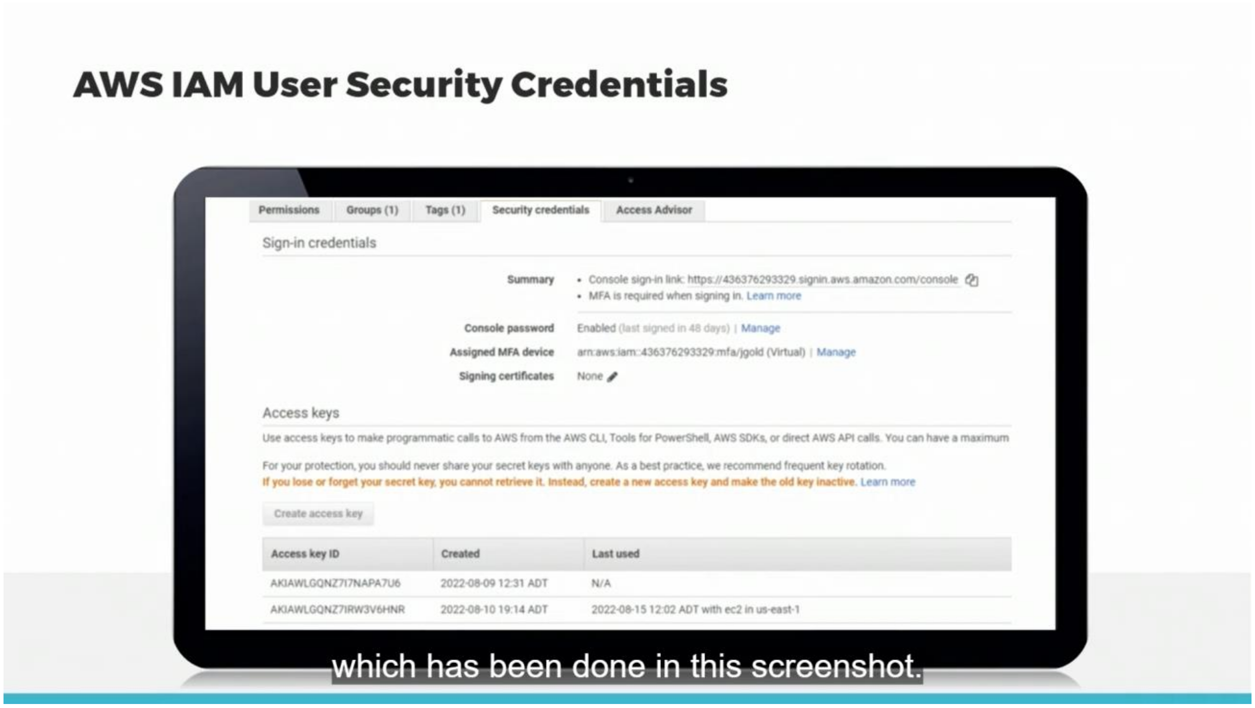
1. Users
2. Groups
3. Roles

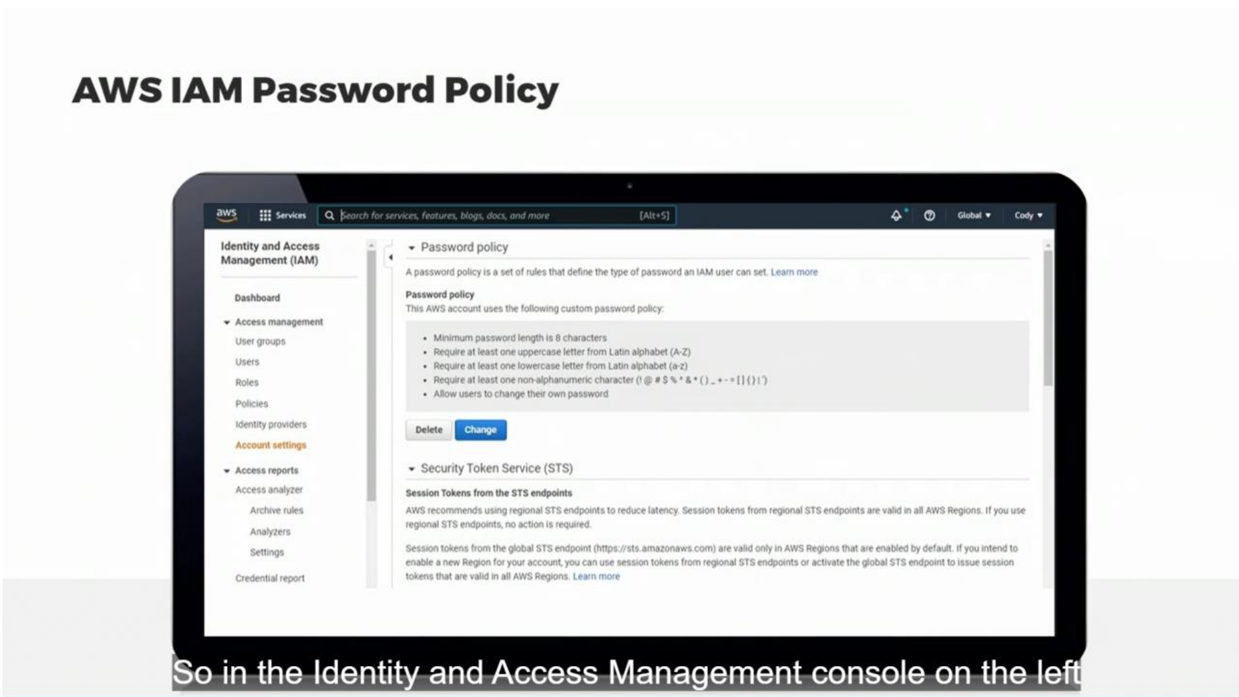
Authentication and Authorization




Built-in AWS Policies

Policy name
AmazonEC2FullAccess
AmazonEC2ReadOnlyAccess
AmazonS3FullAccess
AmazonS3ReadOnlyAccess
AmazonRoute53DomainsFullAccess
AutoScalingFullAccess

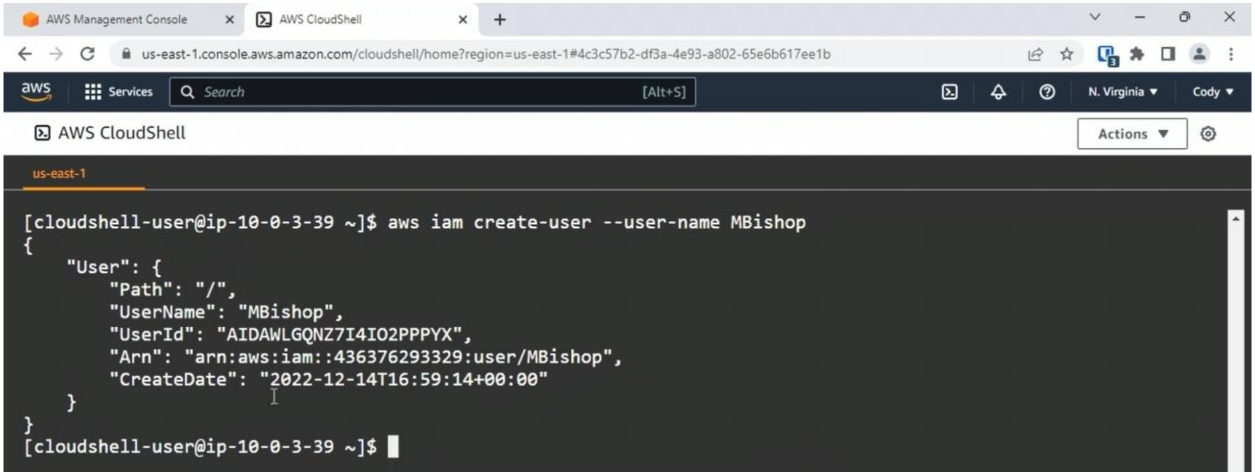




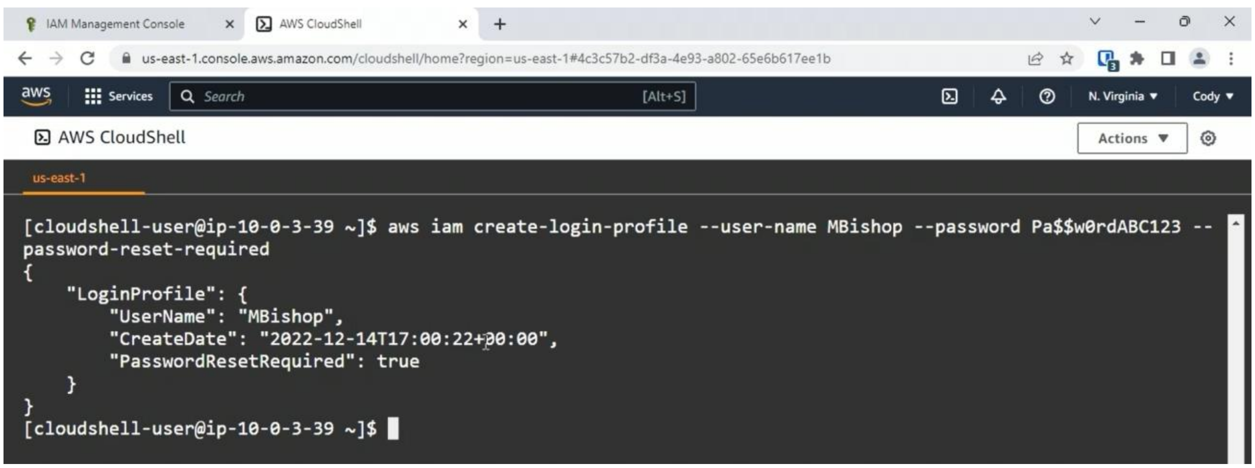
Managing the IAM Users using GUI

Managing the IAM Users using CLI

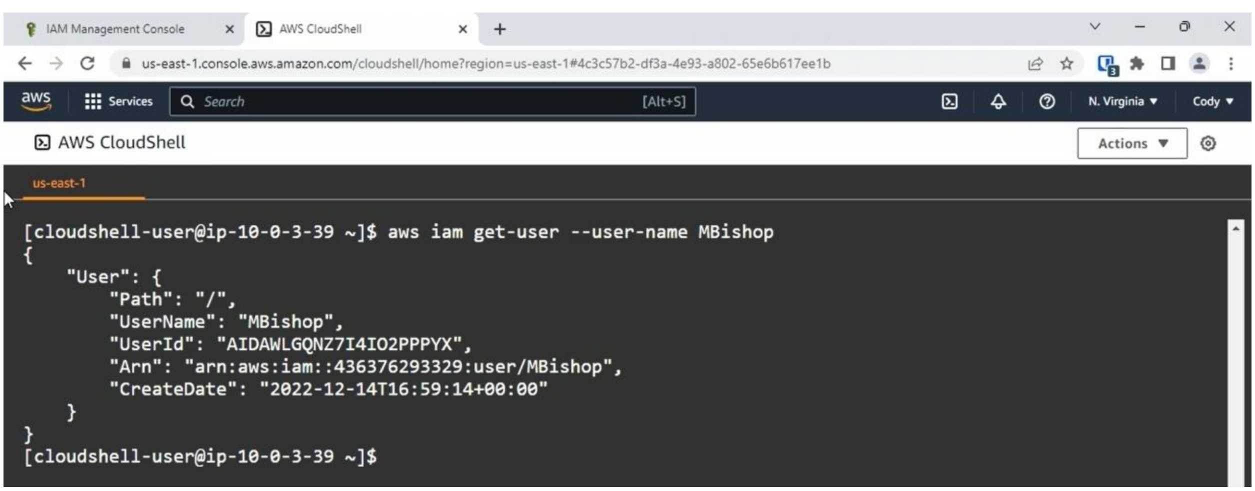
Create user



set password for the user



Get user information



Reset password

```
[cloudshell-user@ip-10-0-3-39 ~]$ aws iam update-login-profile --user-name MBishop --password My\!User1ADiffere
ntP@ssword
[cloudshell-user@ip-10-0-3-39 ~]$
```

Add user to a group

```
[cloudshell-user@ip-10-0-3-39 ~]$ aws iam add-user-to-group --user-name MBishop --group-name East-Developers
[cloudshell-user@ip-10-0-3-39 ~]$ aws iam get-user --user-name MBishop
{
    "User": {
        "Path": "/",
        "UserName": "MBishop",
        "UserId": "AIDAWLGQNZ7I4IO2PPPYX",
        "Arn": "arn:aws:iam::436376293329:user/MBishop",
        "CreateDate": "2022-12-14T16:59:14+00:00",
        "PasswordLastUsed": "2022-12-14T17:02:15+00:00"
    }
}
[cloudshell-user@ip-10-0-3-39 ~]$
```

To check the group of the user

```
[cloudshell-user@ip-10-0-3-39 ~]$ aws iam list-groups-for-user --user-name MBishop
{
    "Groups": [
        {
            "Path": "/",
            "GroupName": "East-Developers",
            "GroupId": "AGPAWLGQNZ7IQ7BXJ7QVW",
            "Arn": "arn:aws:iam::436376293329:group/East-Developers",
            "CreateDate": "2022-12-14T17:04:13+00:00"
        }
```

Managing IAM Users Using PowerShell

Administrator: Windows PowerShell

```
PS C:\> get-awscredential -listprofiledetail

ProfileName StoreTypeName         ProfileLocation
----------- -------------         ---------------
admin1      NetSDKCredentialsFile
default     NetSDKCredentialsFile


PS C:\>
```

Administrator: Windows PowerShell

```
PS C:\> initialize-awsdefaultconfiguration -profilename admin1
PS C:\> get-command *iamuser*

CommandType     Name                                               Version    Source
-----------     ----                                               -------    ------
Alias           Get-IAMUserPolicies                                4.1.196    AWSPowerShell
Alias           Get-IAMUsers                                       4.1.196    AWSPowerShell
Cmdlet          Add-IAMUserTag                                     4.1.196    AWSPowerShell
Cmdlet          Add-IAMUserToGroup                                 4.1.196    AWSPowerShell
Cmdlet          Get-IAMUser                                        4.1.196    AWSPowerShell
Cmdlet          Get-IAMUserList                                    4.1.196    AWSPowerShell
Cmdlet          Get-IAMUserPolicy                                  4.1.196    AWSPowerShell
Cmdlet          Get-IAMUserPolicyList                              4.1.196    AWSPowerShell
Cmdlet          Get-IAMUserTagList                                 4.1.196    AWSPowerShell
Cmdlet          New-IAMUser                                        4.1.196    AWSPowerShell
Cmdlet          Register-IAMUserPolicy                             4.1.196    AWSPowerShell
Cmdlet          Remove-IAMUser                                     4.1.196    AWSPowerShell
Cmdlet          Remove-IAMUserFromGroup                            4.1.196    AWSPowerShell
Cmdlet          Remove-IAMUserPermissionsBoundary                  4.1.196    AWSPowerShell
Cmdlet          Remove-IAMUserPolicy                               4.1.196    AWSPowerShell
Cmdlet          Remove-IAMUserTag                                  4.1.196    AWSPowerShell
Cmdlet          Set-IAMUserPermissionsBoundary                     4.1.196    AWSPowerShell
Cmdlet          Unregister-IAMUserPolicy                           4.1.196    AWSPowerShell
```

Select Administrator: Windows PowerShell

```
PS C:\> get-iamuser


Arn                 : arn:aws:iam::436376293329:root
CreateDate          : 7/3/2018 6:07:58 PM
PasswordLastUsed    : 12/14/2022 3:56:19 PM
Path                :
PermissionsBoundary :
Tags                : {}
UserId              : 436376293329
UserName            :



PS C:\>
```

iamuser – root account

iamusers – all the users

Select Administrator: Windows PowerShell

```
PS C:\> get-iamusers


Arn                 : arn:aws:iam::436376293329:user/JGold
CreateDate          : 12/14/2022 4:08:57 PM
PasswordLastUsed    : 12/14/2022 4:13:55 PM
Path                : /
PermissionsBoundary :
Tags                : {}
UserId              : AIDAWLGQNZ7I7XCLPZNVC
UserName            : JGold

Arn                 : arn:aws:iam::436376293329:user/MBishop
CreateDate          : 12/14/2022 4:59:14 PM
PasswordLastUsed    : 12/14/2022 5:02:15 PM
Path                : /
PermissionsBoundary :
Tags                : {}
UserId              : AIDAWLGQNZ7I4IO2PPPYX
UserName            : MBishop
```

Select Administrator: Windows PowerShell

```
PS C:\> get-command *iamuser*

CommandType     Name                                    Version    Source
-----------     ----                                    -------    ------
Alias           Get-IAMUserPolicies                     4.1.196    AWSPowerShell
Alias           Get-IAMUsers                            4.1.196    AWSPowerShell
Cmdlet          Add-IAMUserTag                          4.1.196    AWSPowerShell
Cmdlet          Add-IAMUserToGroup                      4.1.196    AWSPowerShell
Cmdlet          Get-IAMUser                             4.1.196    AWSPowerShell
Cmdlet          Get-IAMUserList                         4.1.196    AWSPowerShell
Cmdlet          Get-IAMUserPolicy                       4.1.196    AWSPowerShell
Cmdlet          Get-IAMUserPolicyList                   4.1.196    AWSPowerShell
Cmdlet          Get-IAMUserTagList                      4.1.196    AWSPowerShell
Cmdlet          New-IAMUser                             4.1.196    AWSPowerShell
Cmdlet          Register-IAMUserPolicy                  4.1.196    AWSPowerShell
Cmdlet          Remove-IAMUser                          4.1.196    AWSPowerShell
Cmdlet          Remove-IAMUserFromGroup                 4.1.196    AWSPowerShell
Cmdlet          Remove-IAMUserPermissionsBoundary       4.1.196    AWSPowerShell
Cmdlet          Remove-IAMUserPolicy                    4.1.196    AWSPowerShell
Cmdlet          Remove-IAMUserTag                       4.1.196    AWSPowerShell
Cmdlet          Set-IAMUserPermissionsBoundary          4.1.196    AWSPowerShell
Cmdlet          Unregister-IAMUserPolicy                4.1.196    AWSPowerShell
Cmdlet          Update-IAMUser                          4.1.196    AWSPowerShell
Cmdlet          Write-IAMUserPolicy                     4.1.196    AWSPowerShell


PS C:\> _
```

We can easily reset a user password in the CLI

Administrator: Windows PowerShell

```
PS C:\> new-iamuser -username AAdachi


Arn                 : arn:aws:iam::436376293329:user/AAdachi
CreateDate          : 12/14/2022 5:29:59 PM
PasswordLastUsed    : 1/1/0001 12:00:00 AM
Path                : /
PermissionsBoundary :
Tags                : {}
UserId              : AIDAWLGQNZ7IYALFMPQNU
UserName            : AAdachi


PS C:\> get-iamusers_
```

Select Administrator: Windows PowerShell

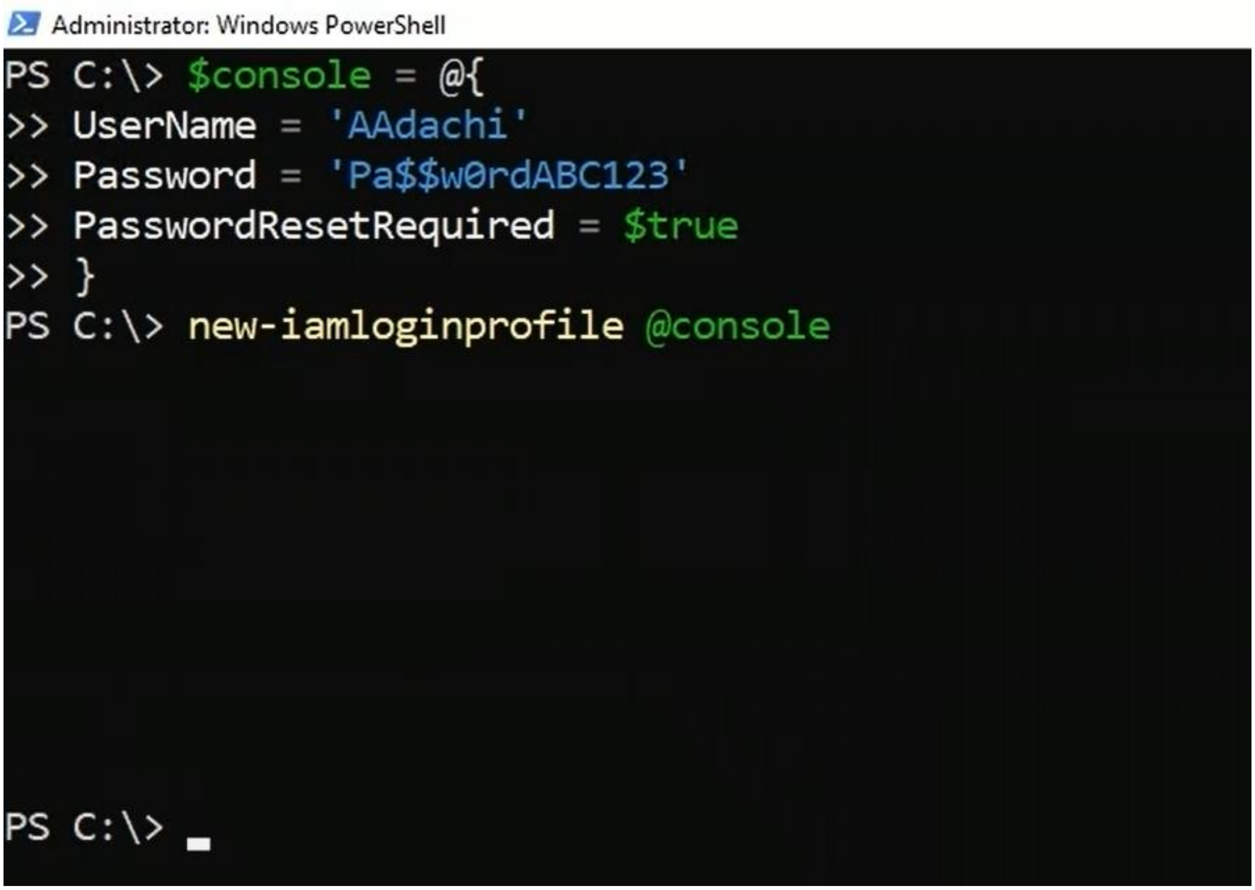
```
PS C:\> get-iamgroups


Arn        : arn:aws:iam::436376293329:group/East-Developers
CreateDate : 12/14/2022 5:04:13 PM
GroupId    : AGPAWLGQNZ7IQ7BXJ7QVW
GroupName  : East-Developers
Path       : /



PS C:\>
```

Administrator: Windows PowerShell

```
PS C:\> add-iamusertogroup -username aadachi -groupname east-developers
PS C:\>
```

Select Administrator: Windows PowerShell

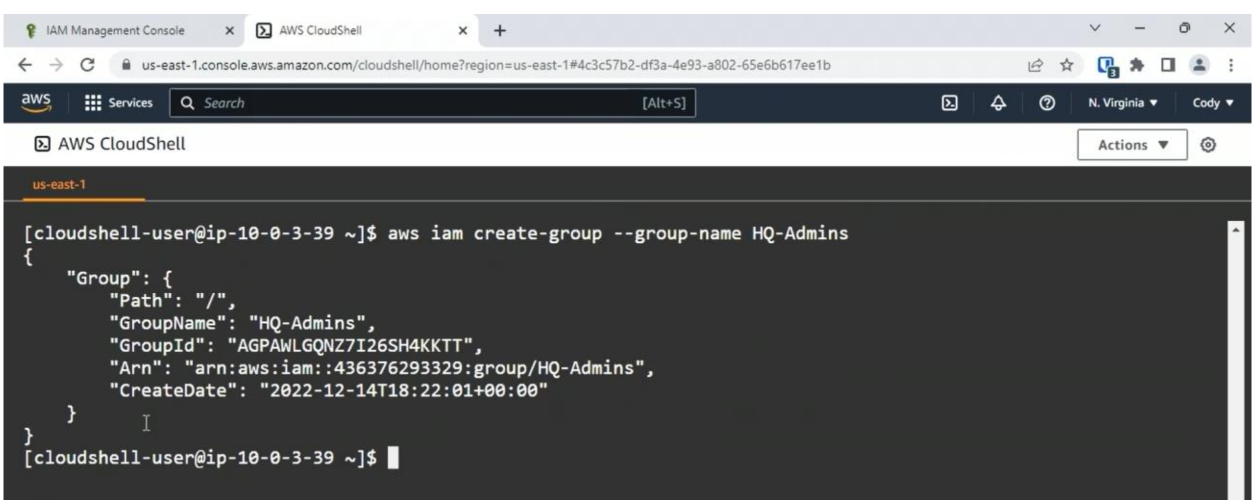
```
PS C:\> get-iamgroup -groupname east-developers

Group                              IsTruncated Marker Users
-----                              ----------- ------ -----
Amazon.IdentityManagement.Model.Group False             {MBishop, AAdachi}


PS C:\>
```



Managing IAM Groups Using the GUI

Managing IAM Groups Using the CLI

Aws iam help



aws iam list-groups – to get list of groups

```
[cloudshell-user@ip-10-0-3-39 ~]$ aws iam list-groups --query "Groups[].Groupname"
[]
[cloudshell-user@ip-10-0-3-39 ~]$ aws iam list-groups --query "Groups[].GroupName"
[
    "East-Developers",
    "HQ-Admins",
    "West-Developers"
]
[cloudshell-user@ip-10-0-3-39 ~]$
```

```
[cloudshell-user@ip-10-0-3-39 ~]$ aws iam get-group --group hq-admins
{
    "Users": [],
    "Group": {
        "Path": "/",
        "GroupName": "HQ-Admins",
        "GroupId": "AGPAWLGQNZ7I26SH4KKTT",
        "Arn": "arn:aws:iam::436376293329:group/HQ-Admins",
        "CreateDate": "2022-12-14T18:22:01+00:00"
    }
}
[cloudshell-user@ip-10-0-3-39 ~]$
```

Add and remove user from group

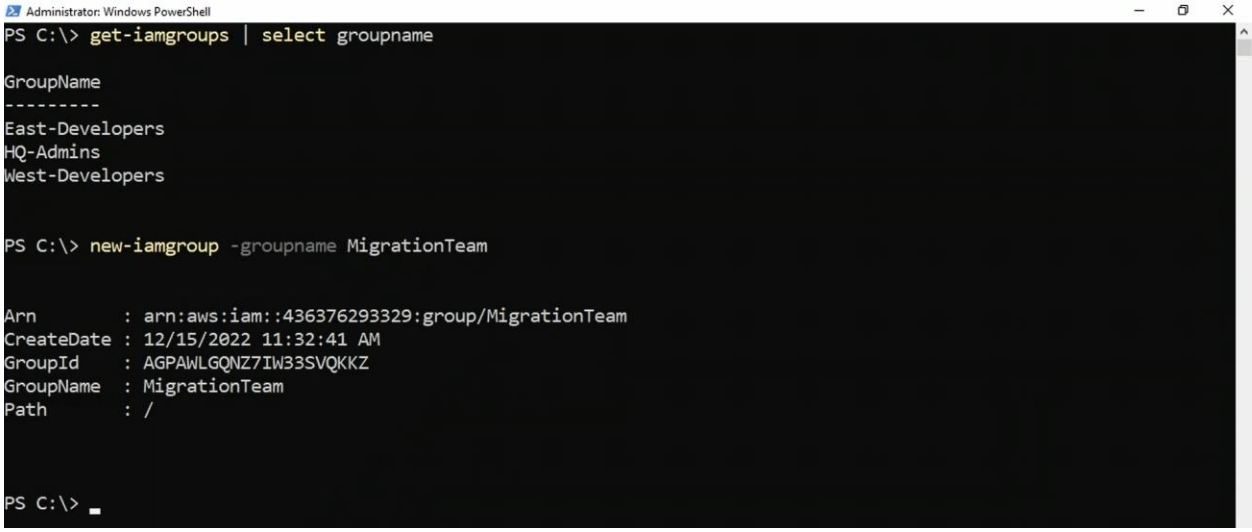
```
[cloudshell-user@ip-10-0-3-39 ~]$ aws iam add-user-to-group --user-name JGold --group-name hq-admins
[cloudshell-user@ip-10-0-3-39 ~]$ aws iam remove-user-from-group --group-name hq-admins --user-name jgold
[cloudshell-user@ip-10-0-3-39 ~]$
```

Managing IAM Groups Using PowerShell

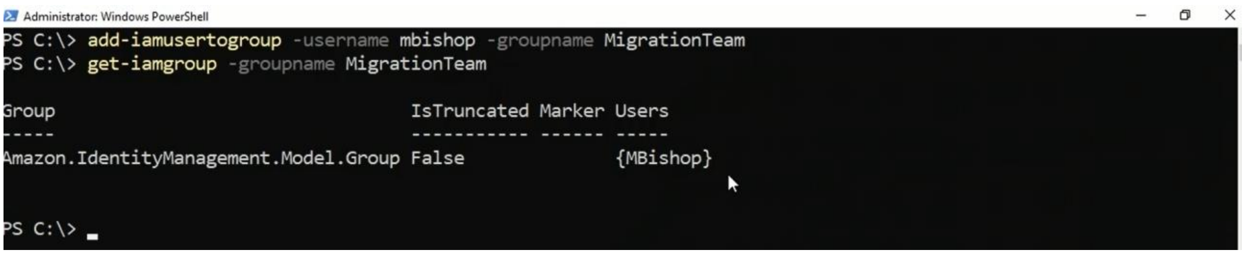
```
Administrator: Windows PowerShell
PS C:\> get-command *iamgroup*

CommandType     Name                                               Version    Source
-----------     ----                                               -------    ------
Alias           Get-IAMGroupPolicies                               4.1.196    AWSPowerShell
Alias           Get-IAMGroups                                      4.1.196    AWSPowerShell
Cmdlet          Get-IAMGroup                                       4.1.196    AWSPowerShell
Cmdlet          Get-IAMGroupForUser                                4.1.196    AWSPowerShell
Cmdlet          Get-IAMGroupList                                   4.1.196    AWSPowerShell
Cmdlet          Get-IAMGroupPolicy                                 4.1.196    AWSPowerShell
Cmdlet          Get-IAMGroupPolicyList                             4.1.196    AWSPowerShell
Cmdlet          New-IAMGroup                                       4.1.196    AWSPowerShell
Cmdlet          Register-IAMGroupPolicy                            4.1.196    AWSPowerShell
Cmdlet          Remove-IAMGroup                                    4.1.196    AWSPowerShell
Cmdlet          Remove-IAMGroupPolicy                              4.1.196    AWSPowerShell
Cmdlet          Unregister-IAMGroupPolicy                          4.1.196    AWSPowerShell
Cmdlet          Update-IAMGroup                                    4.1.196    AWSPowerShell
Cmdlet          Write-IAMGroupPolicy                               4.1.196    AWSPowerShell


PS C:\>
```

```
Administrator: Windows PowerShell
PS C:\> get-iamgroup -groupname east-developers

Group                              IsTruncated Marker Users
-----                              ----------- ------ -----
Amazon.IdentityManagement.Model.Group False              {MBishop, AAdachi}


PS C:\>
```

```
PS C:\> get-iamgroups


Arn        : arn:aws:iam::436376293329:group/East-Developers
CreateDate : 12/14/2022 5:04:13 PM
GroupId    : AGPAWLGQNZ7IQ7BX[illegible]7QVW
GroupName  : East-Developers
Path       : /

Arn        : arn:aws:iam::436376293329:group/HQ-Admins
CreateDate : 12/14/2022 6:22:01 PM
GroupId    : AGPAWLGQNZ7I26SH4KKTT
GroupName  : HQ-Admins
Path       : /

Arn        : arn:aws:iam::436376293329:group/West-Developers
CreateDate : 12/14/2022 6:05:46 PM
GroupId    : AGPAWLGQNZ7I2DYA5AE4L
GroupName  : West-Developers
Path       : /
```

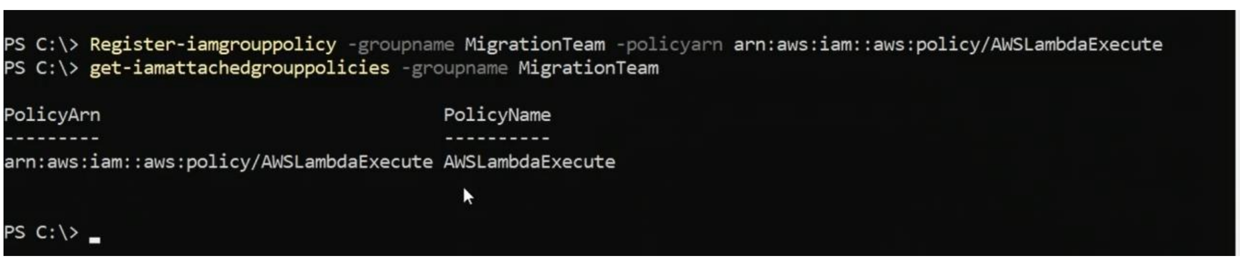


```
Administrator: Windows PowerShell
PS C:\> get-iamgroups | select groupname

GroupName
---------
East-Developers
HQ-Admins
West-Developers


PS C:\> new-iamgroup -groupname MigrationTeam


Arn        : arn:aws:iam::436376293329:group/MigrationTeam
CreateDate : 12/15/2022 11:32:41 AM
GroupId    : AGPAWLGQNZ7IW33SVQKKZ
GroupName  : MigrationTeam
Path       : /



PS C:\>
```



```
Administrator: Windows PowerShell
PS C:\> add-iamusertogroup -username mbishop -groupname MigrationTeam
PS C:\> get-iamgroup -groupname MigrationTeam

Group                              IsTruncated Marker Users
-----                              ----------- ------ -----
Amazon.IdentityManagement.Model.Group False              {MBishop}


PS C:\>
```

Attach policy

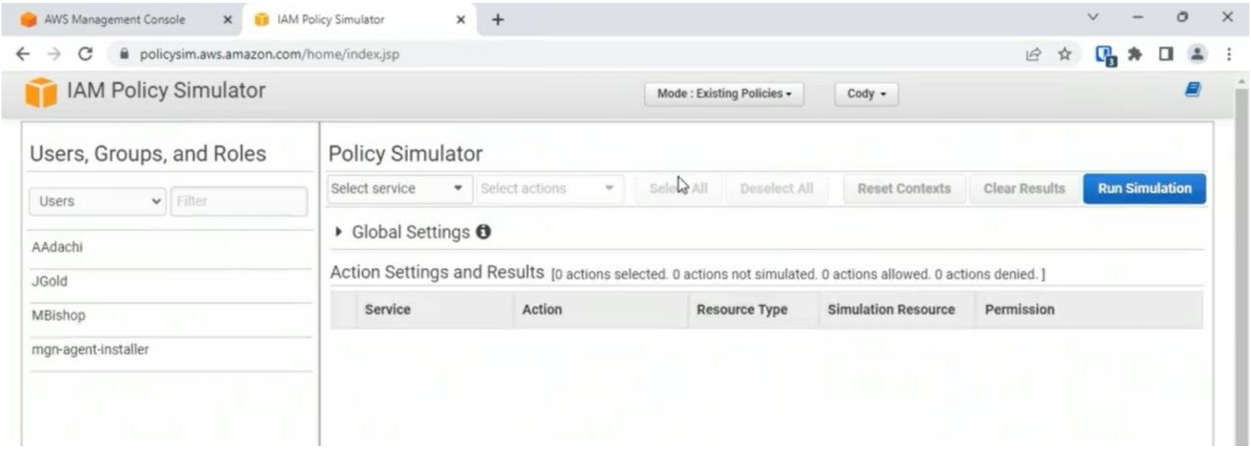


```
PS C:\> Register-iamgrouppolicy -groupname MigrationTeam -policyarn arn:aws:iam::aws:policy/AWSLambdaExecute
PS C:\> get-iamattachedgrouppolicies -groupname MigrationTeam

PolicyArn                              PolicyName
---------                              ----------
arn:aws:iam::aws:policy/AWSLambdaExecute AWSLambdaExecute


PS C:\>
```

Configuring IAM policies

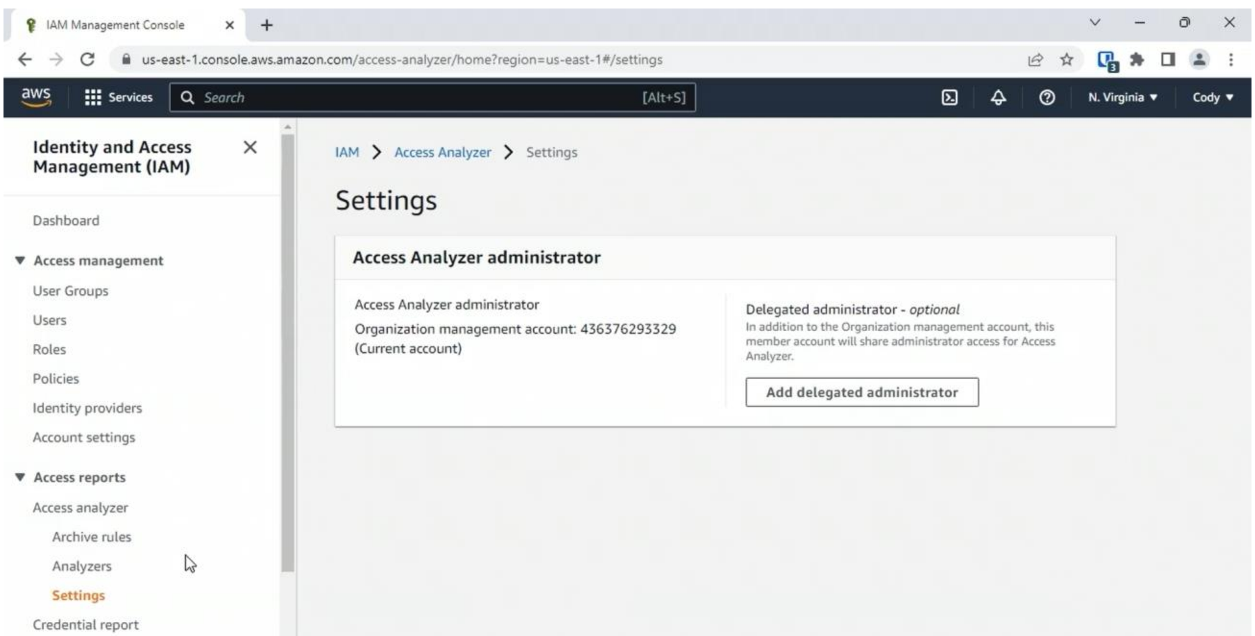
Configuring IAM Roles

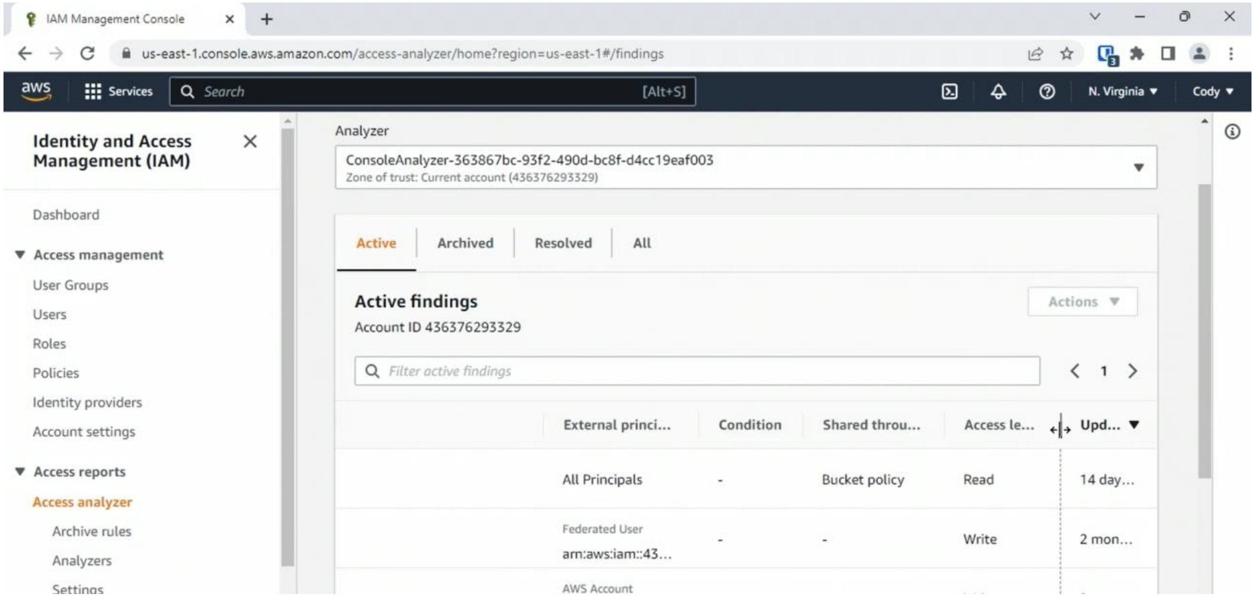
Using the IAM Policy Simulator

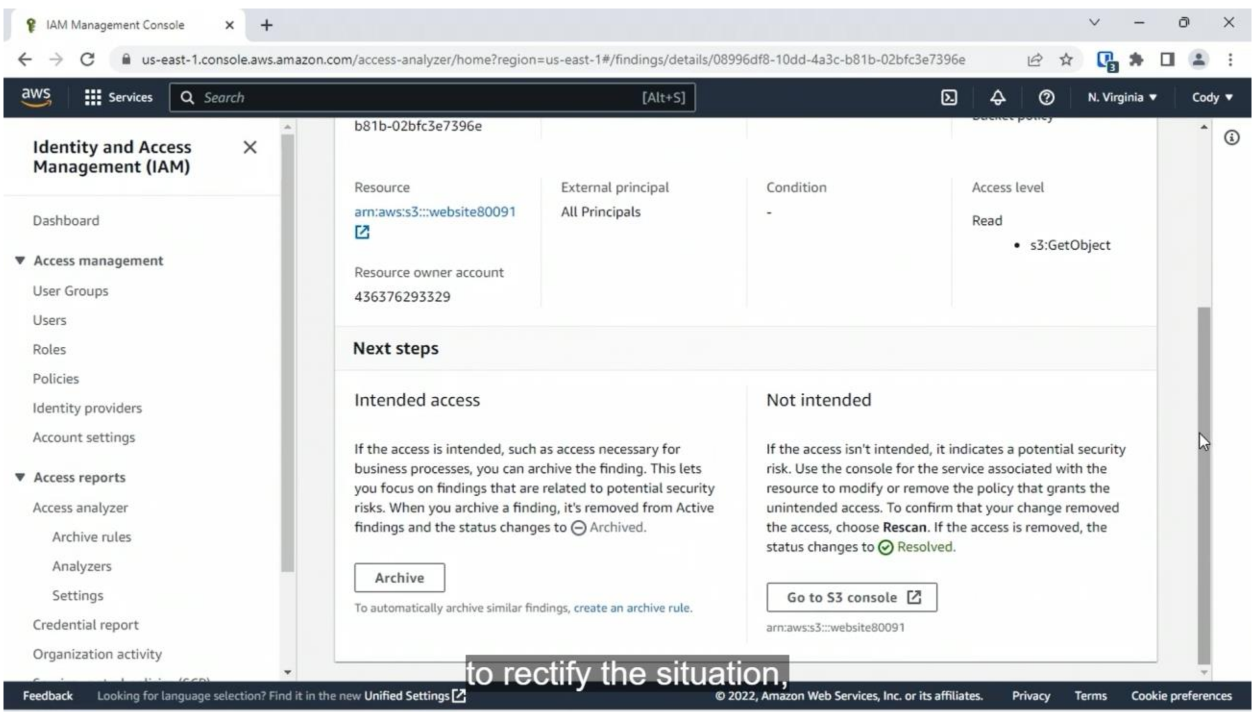


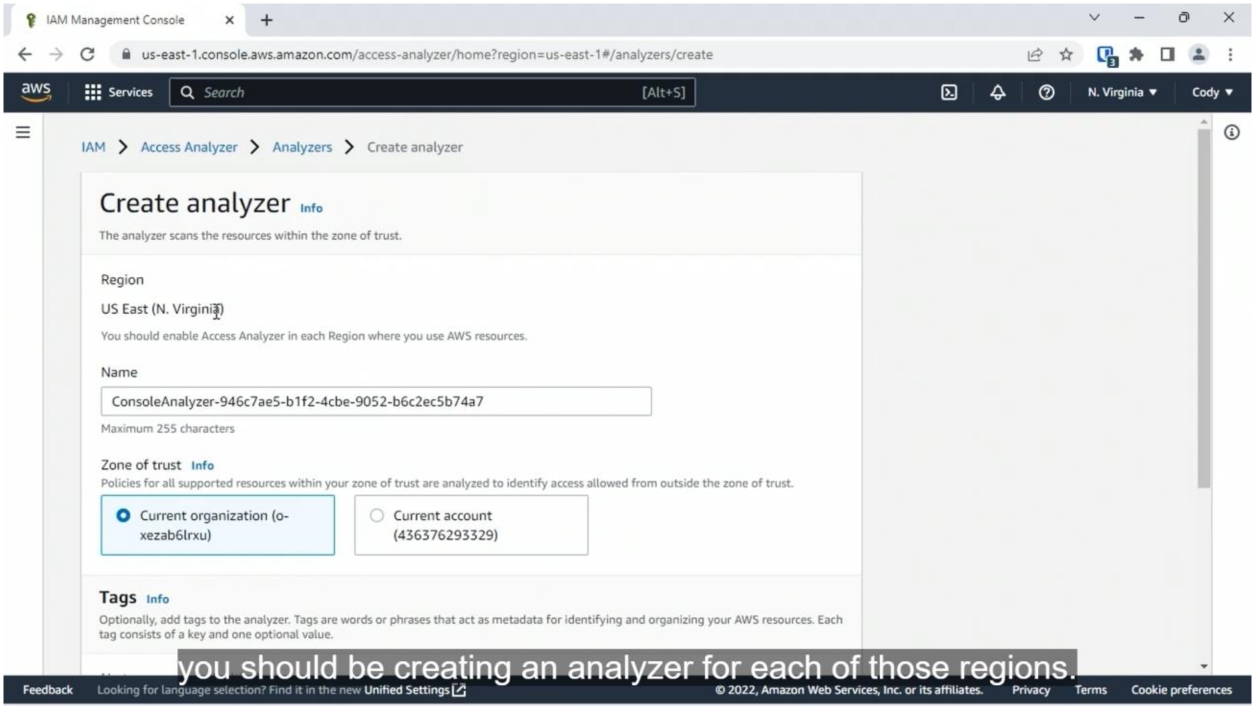
Can select users, groups and roles from the dropdown

Using the IAM Access Analyzer



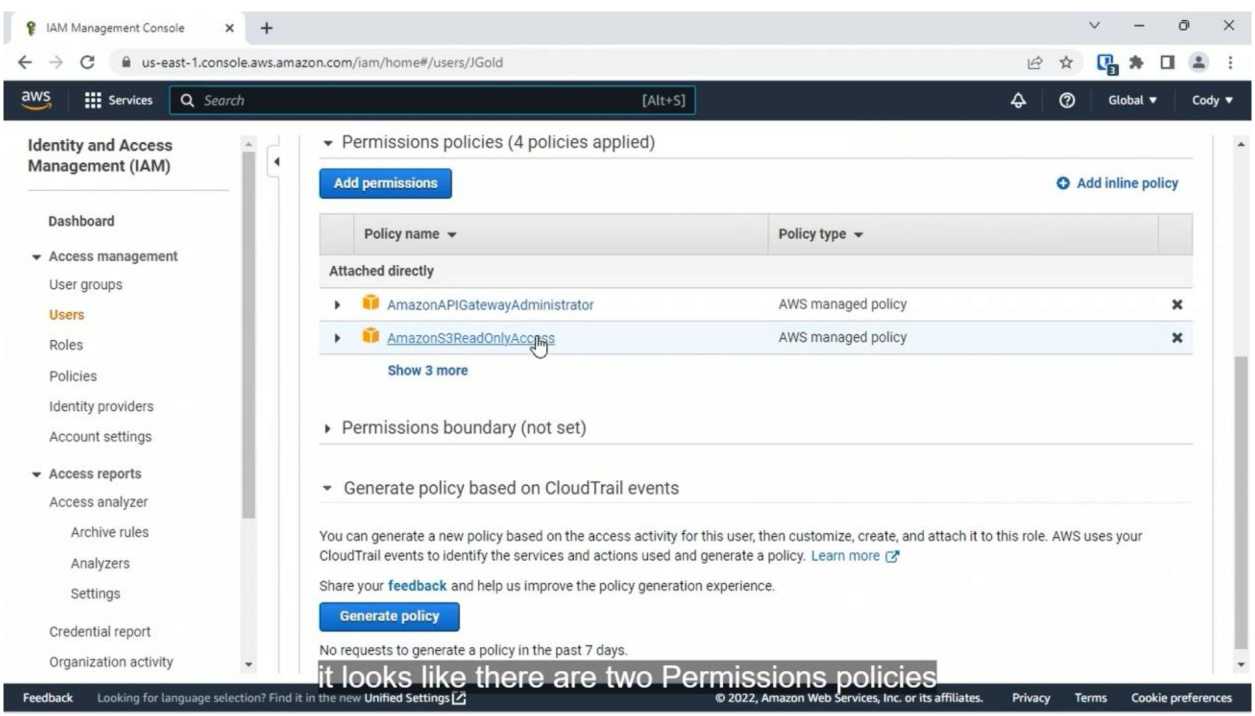

IAM Management Console
us-east-1.console.aws.amazon.com/access-analyzer/home?region=us-east-1#/findings
Services
Search
[Alt+S]
N. Virginia
Cody
Identity and Access Management (IAM)
Dashboard
Access management
User Groups
Users
Roles
Policies
Identity providers
Account settings
Access reports
Access analyzer
Archive rules
Analyzers
Settings
Analyzer
ConsoleAnalyzer-363867bc-93f2-490d-bc8f-d4cc19eaf003
Zone of trust: Current account (436376293329)
Active
Archived
Resolved
All
Active findings
Account ID 436376293329
Actions
Filter active findings
External princi...
Condition
Shared throu...
Access le...
Upd...
All Principals
-
Bucket policy
Read
14 day...
Federated User
arn:aws:iam::43...
-
-
Write
2 mon...
AWS Account

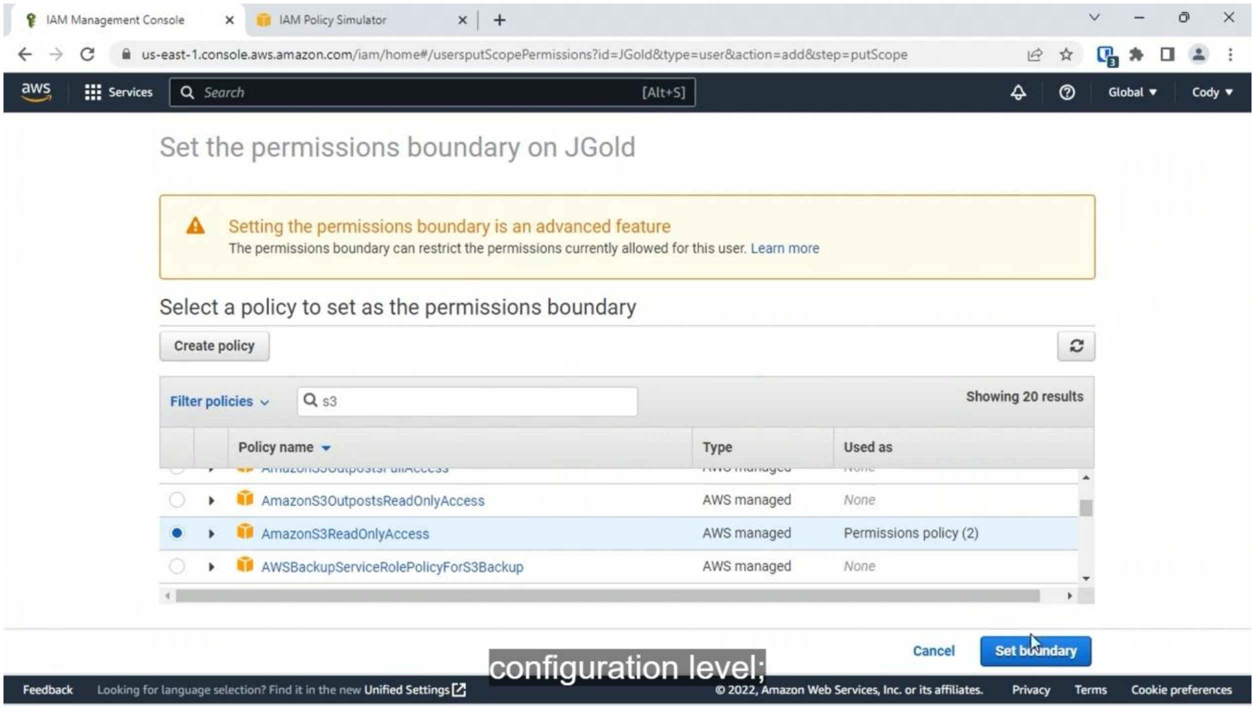

IAM Management Console
us-east-1.console.aws.amazon.com/access-analyzer/home?region=us-east-1#/findings/details/08996df8-10dd-4a3c-b81b-02bfc3e7396e
Services
Search
[Alt+S]
N. Virginia
Cody
Identity and Access Management (IAM)
Dashboard
Access management
User Groups
Users
Roles
Policies
Identity providers
Account settings
Access reports
Access analyzer
Archive rules
Analyzers
Settings
Credential report
Organization activity
b81b-02bfc3e7396e
Resource
arn:aws:s3:::website80091
Resource owner account
436376293329
External principal
All Principals
Condition
-
Access level
Read
s3:GetObject
Next steps
Intended access
If the access is intended, such as access necessary for business processes, you can archive the finding. This lets you focus on findings that are related to potential security risks. When you archive a finding, it's removed from Active findings and the status changes to Archived.
Archive
To automatically archive similar findings, create an archive rule.
Not intended
If the access isn't intended, it indicates a potential security risk. Use the console for the service associated with the resource to modify or remove the policy that grants the unintended access. To confirm that your change removed the access, choose Rescan. If the access is removed, the status changes to Resolved.
Go to S3 console
arn:aws:s3:::website80091
to rectify the situation,
Feedback
Looking for language selection? Find it in the new Unified Settings
© 2022, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences



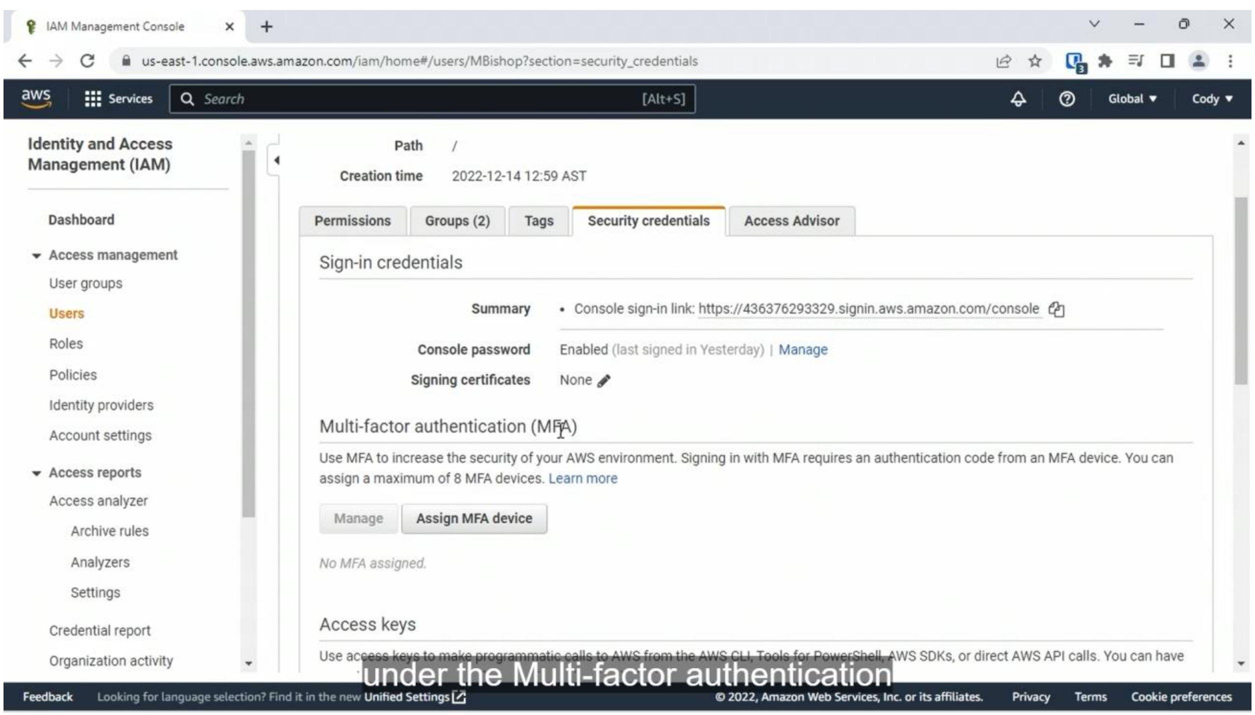
Working with IAM Permissions Boundaries

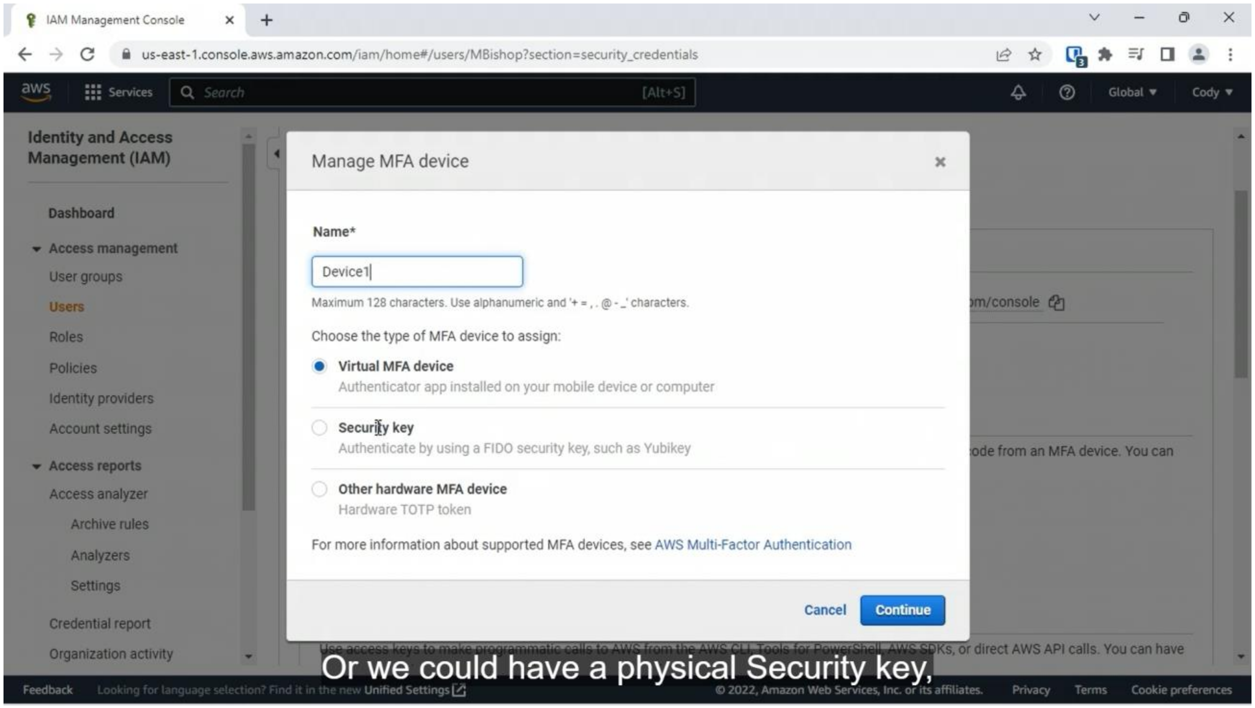
Will find permission boundaries under the individual user or roles and not in the user groups





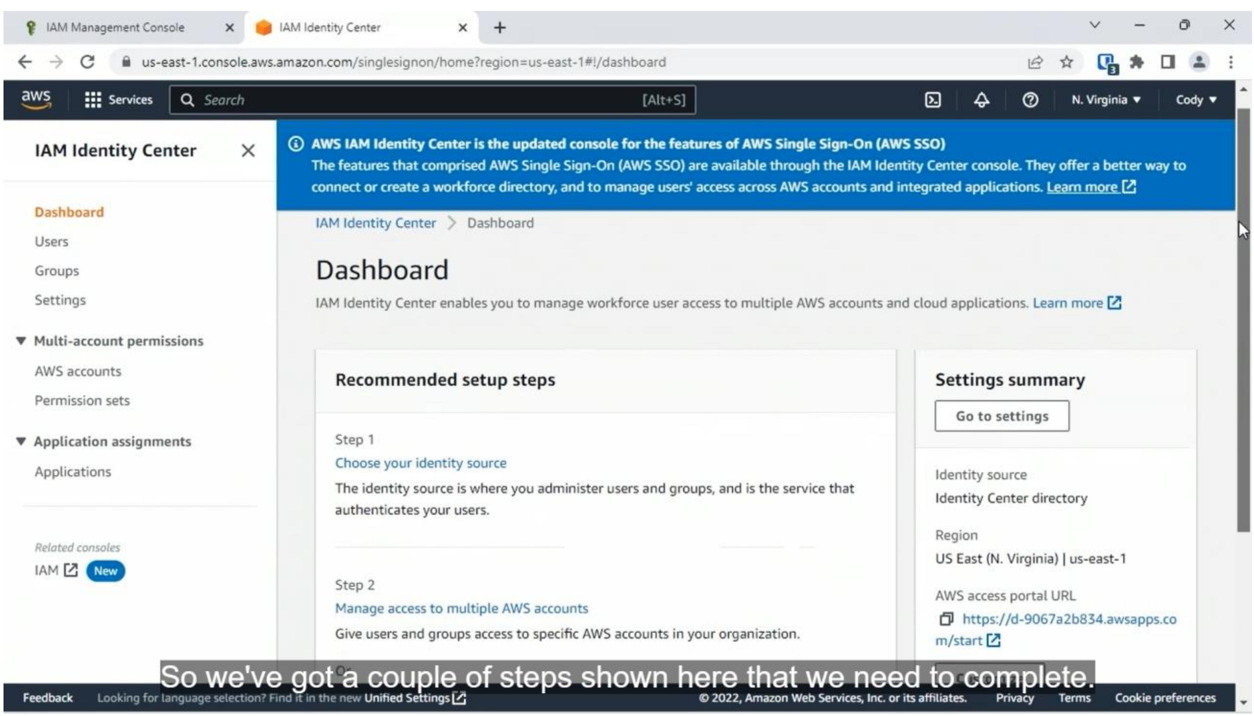
Enabling MFA for an IAM User





Using the IAM Identity Center Console

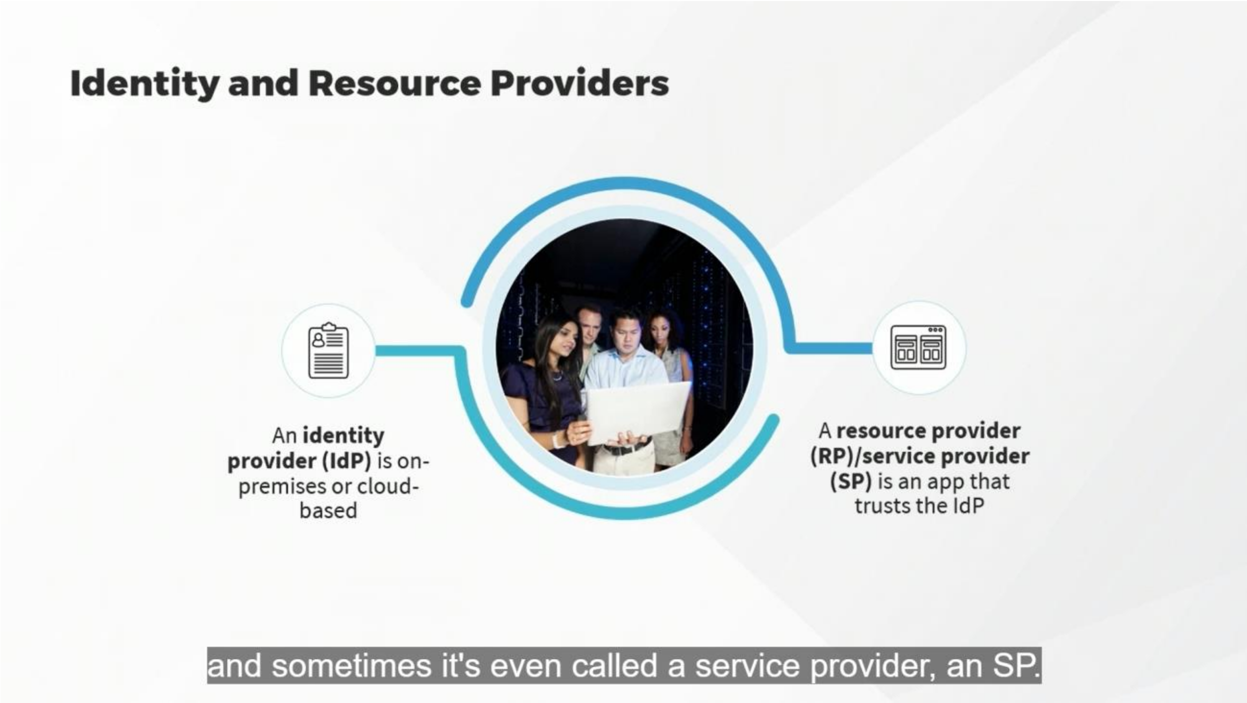
First enable the identity center

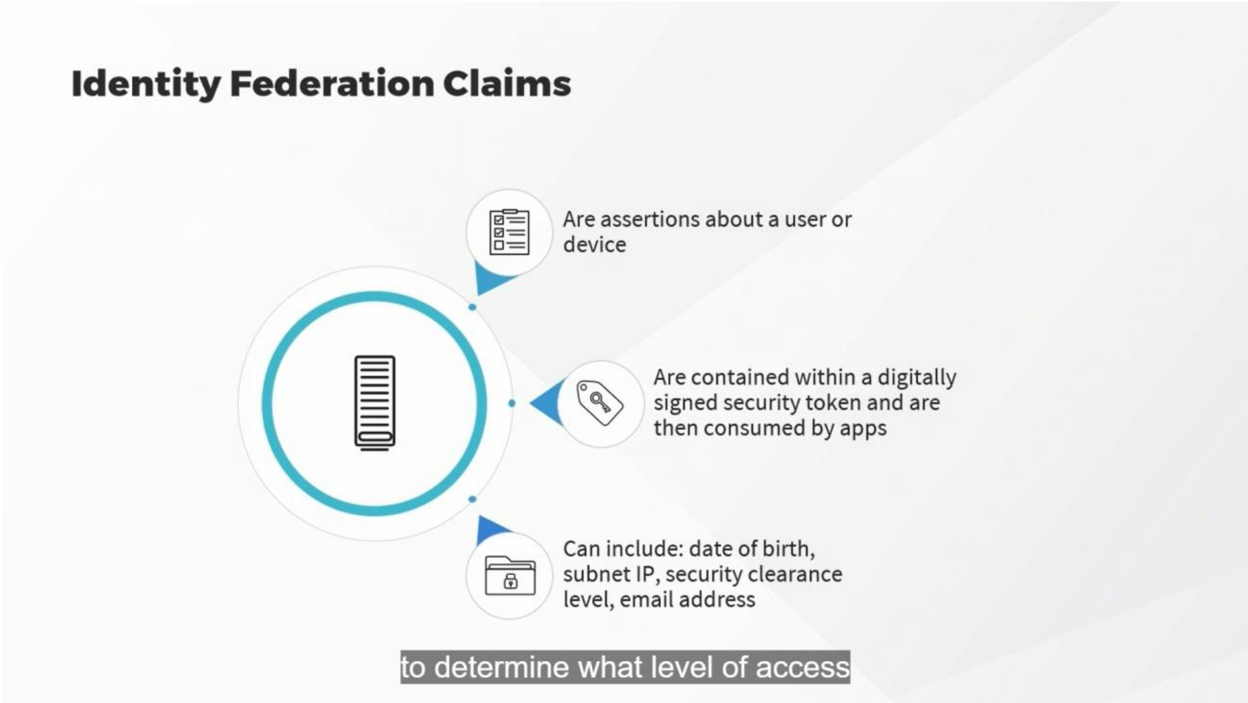


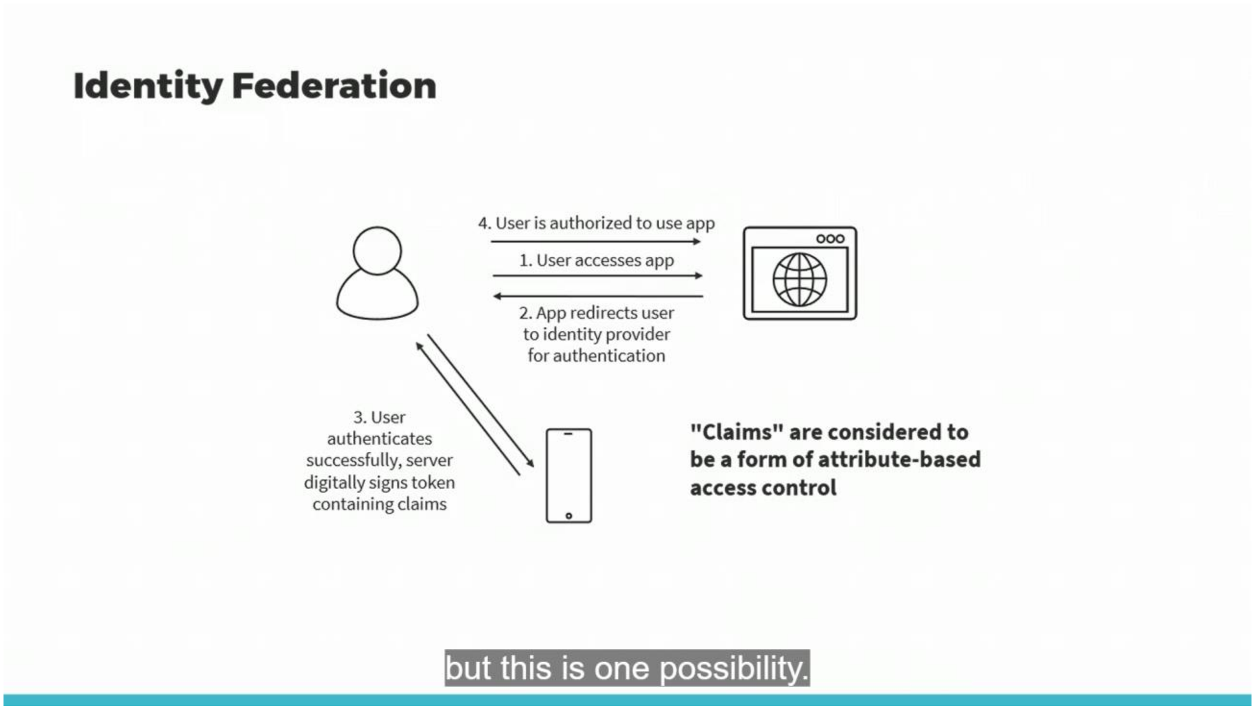
AWS Federated Access

Identity federation

- Uses a centralized trusted authenticator
- Authentication capabilities are removed from apps
- Apps rely upon the trusted authenticator
- A trusted authenticator can span organizations



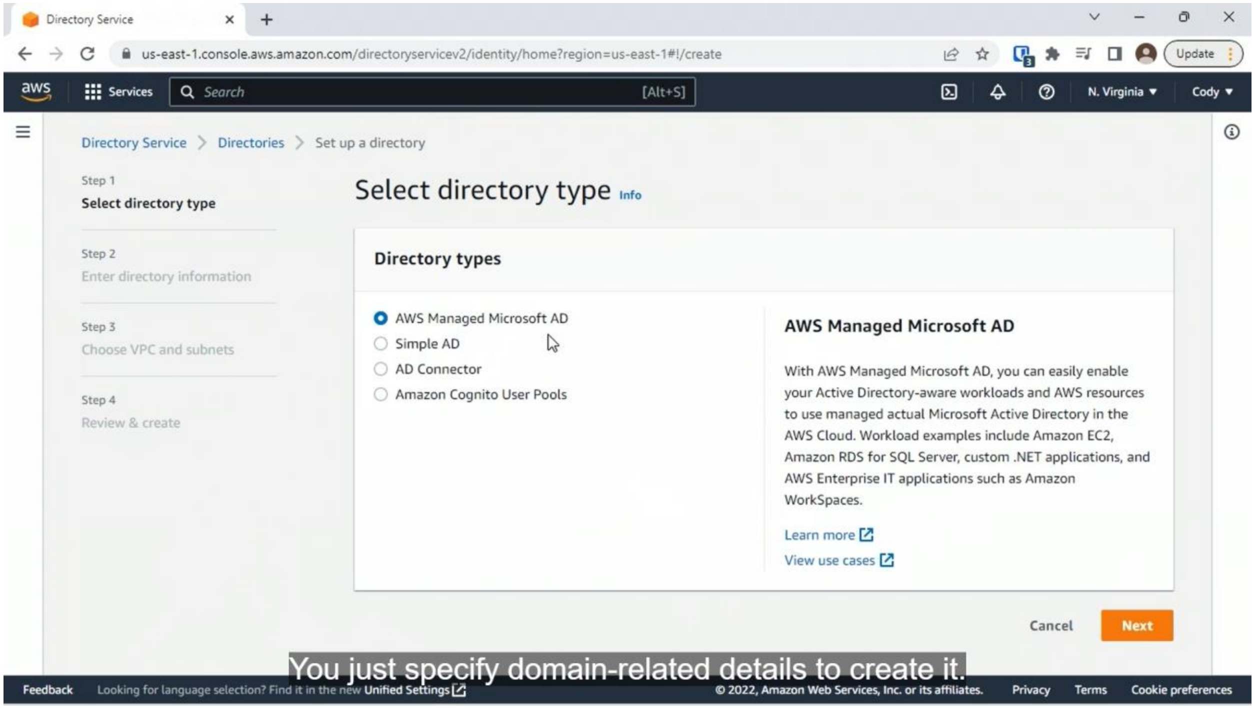


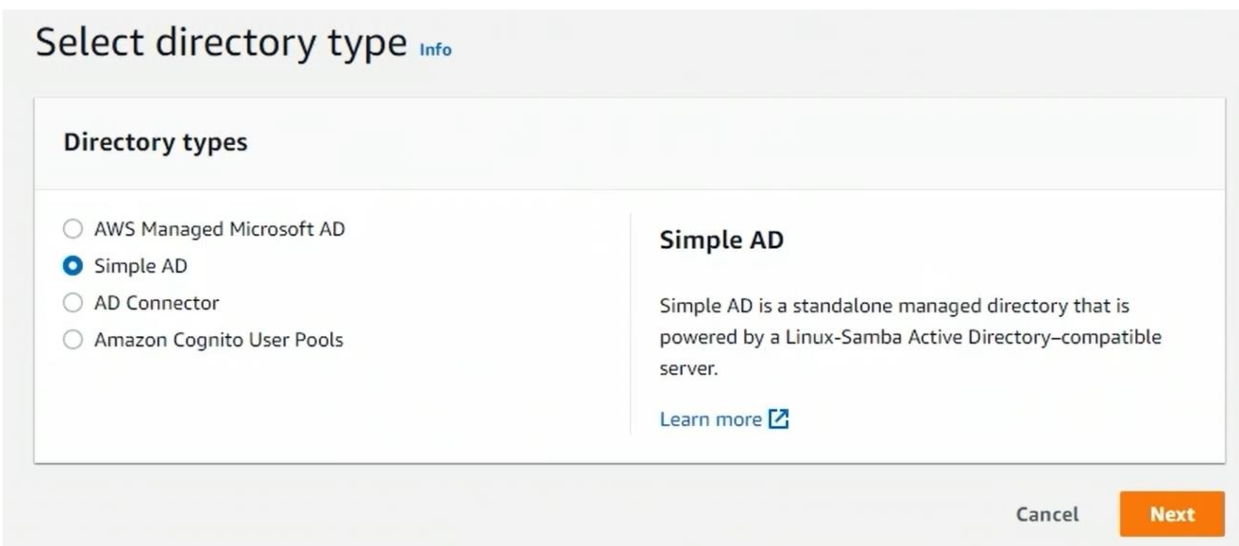


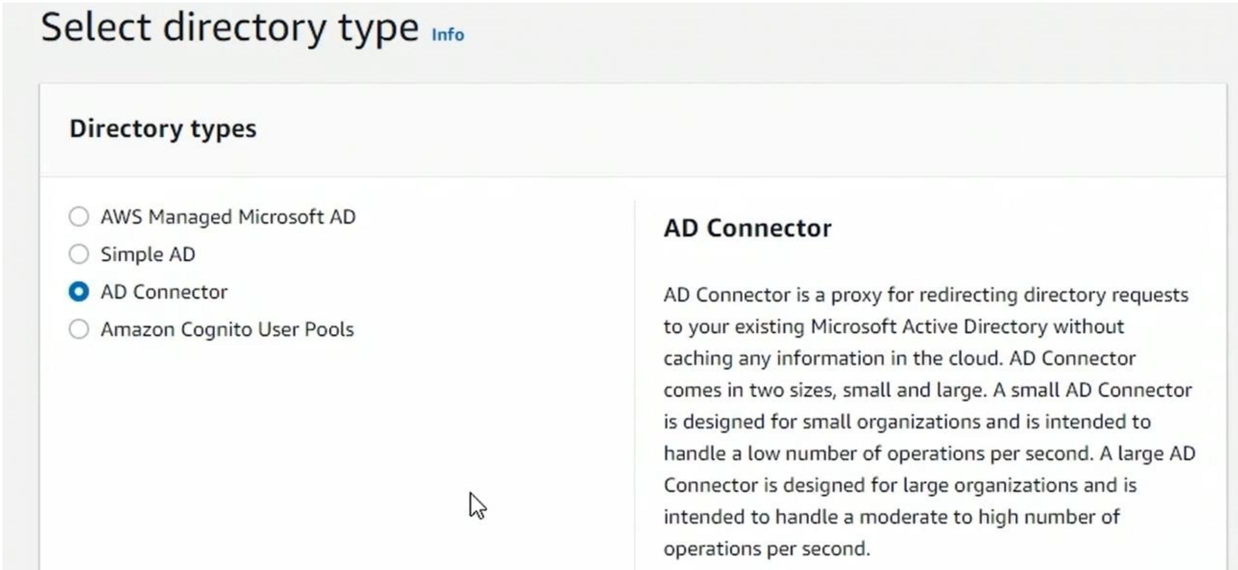
AWS Identity Solutions

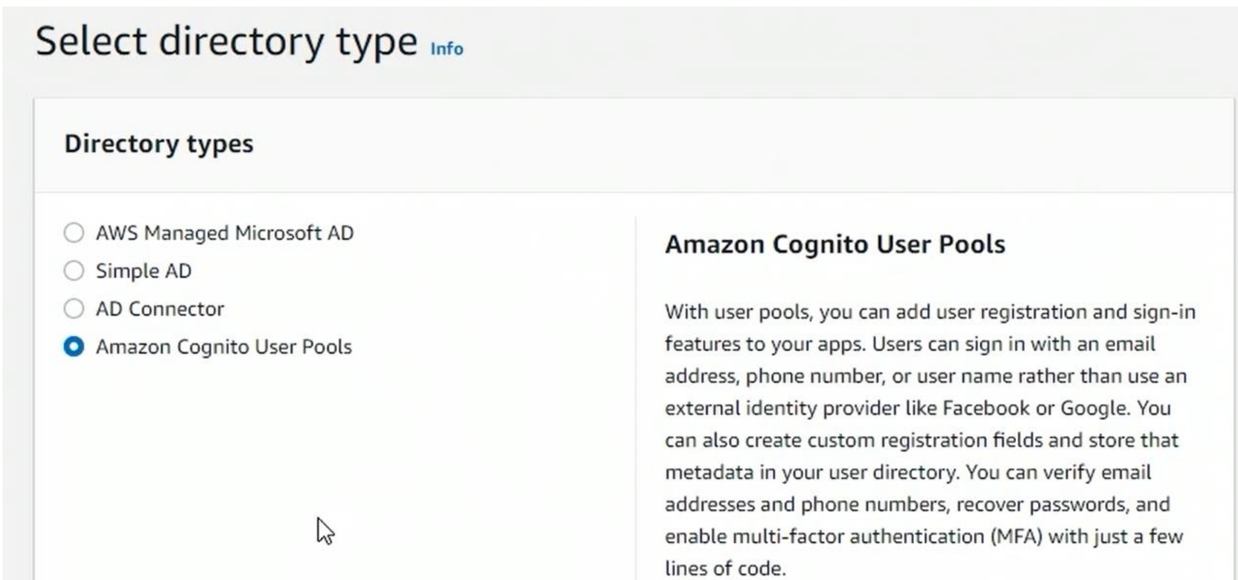
- AWS Single Sign-On (SSO) > AWS IAM Identity Center
- AWS Security Token Service: temporary for like web applications

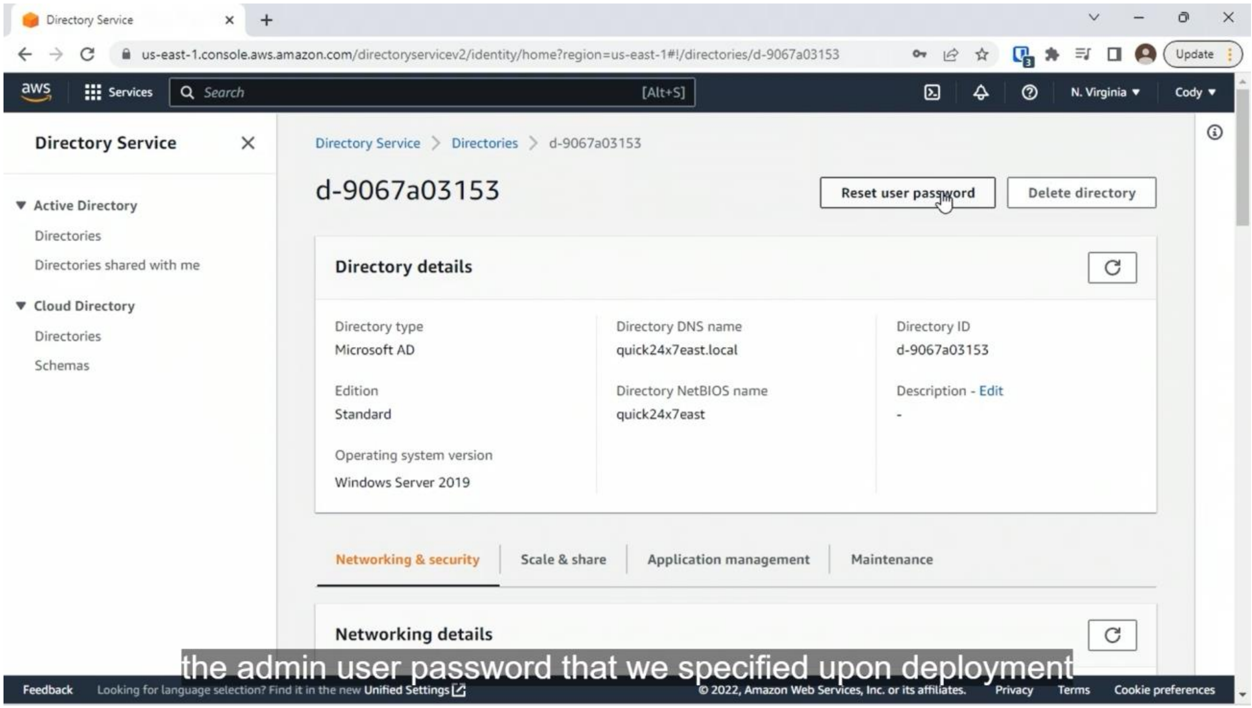
Deploying AWS Managed Active Directory



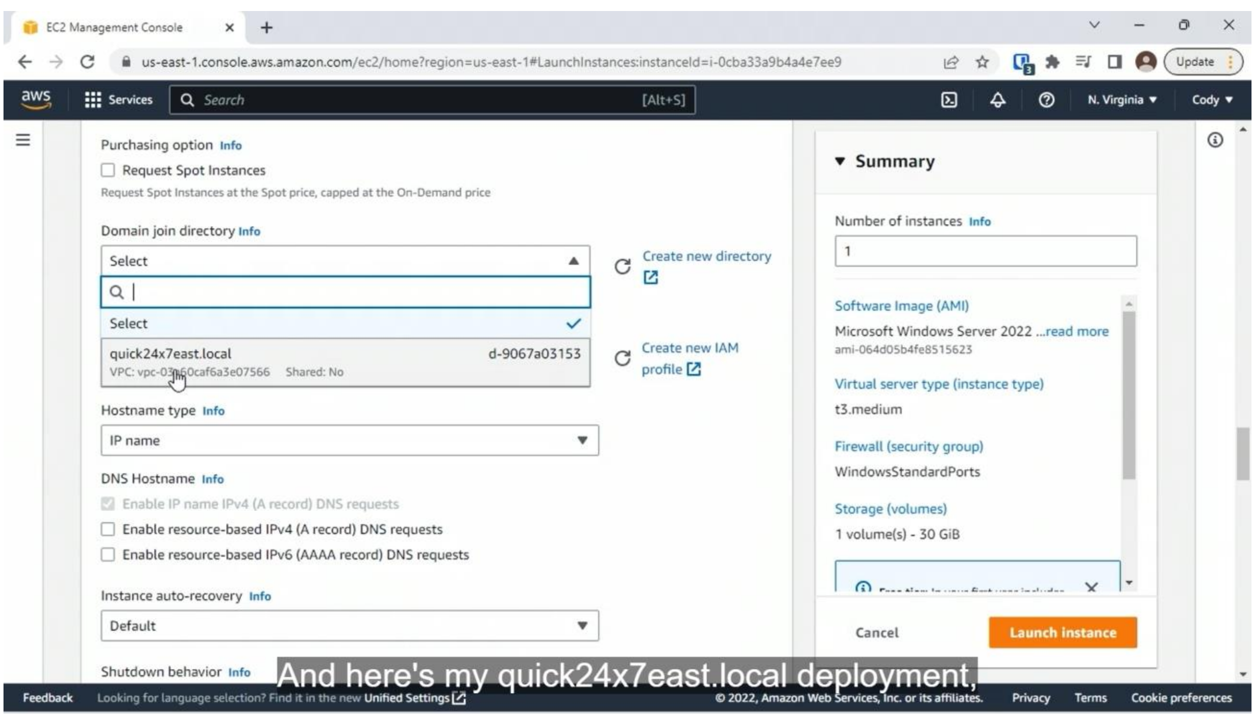


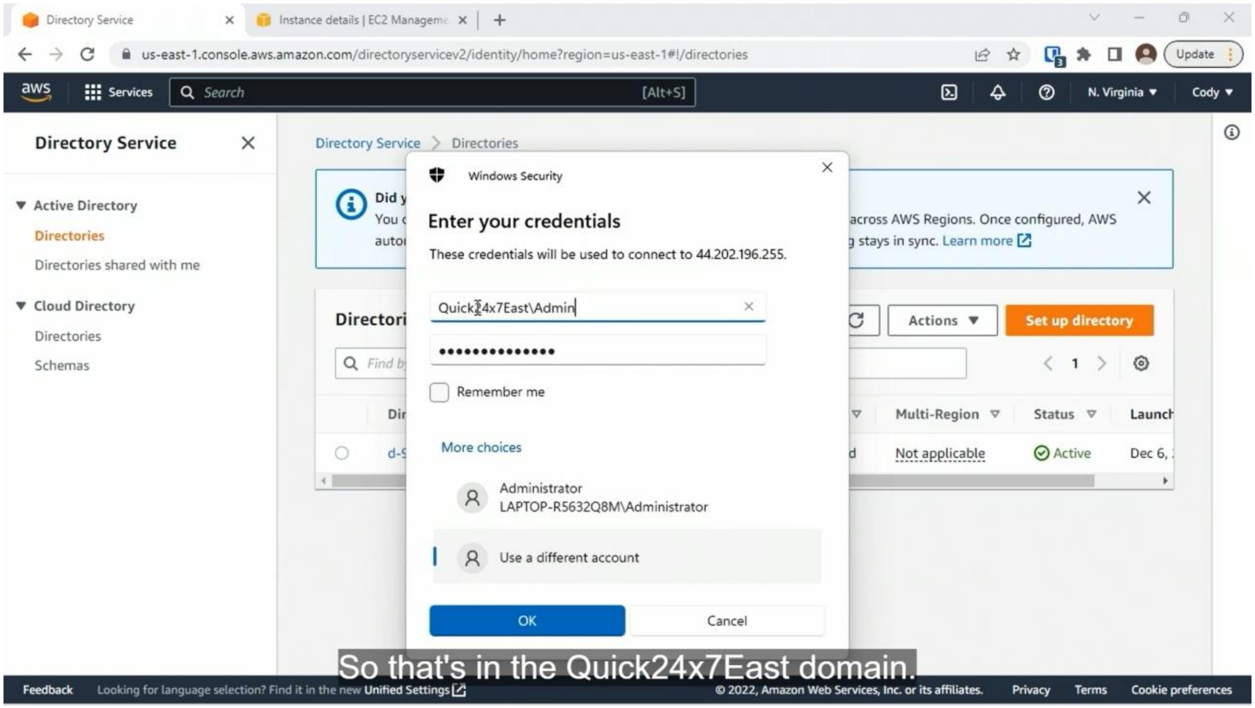

Select directory type Info
Directory types
AWS Managed Microsoft AD
Simple AD
AD Connector
Amazon Cognito User Pools
AD Connector
AD Connector is a proxy for redirecting directory requests to your existing Microsoft Active Directory without caching any information in the cloud. AD Connector comes in two sizes, small and large. A small AD Connector is designed for small organizations and is intended to handle a low number of operations per second. A large AD Connector is designed for large organizations and is intended to handle a moderate to high number of operations per second.


Select directory type Info
Directory types
AWS Managed Microsoft AD
Simple AD
AD Connector
Amazon Cognito User Pools
Amazon Cognito User Pools
With user pools, you can add user registration and sign-in features to your apps. Users can sign in with an email address, phone number, or user name rather than use an external identity provider like Facebook or Google. You can also create custom registration fields and store that metadata in your user directory. You can verify email addresses and phone numbers, recover passwords, and enable multi-factor authentication (MFA) with just a few lines of code.



Joining Stations to AWS Managed Active Directory (AD)





1. When are IAM roles used?
- To assign permissions to a software or service entity

2. What is wrong with the following PowerShell expression?

   Add-IAMUserToGroup -UserName MBishop
- The group name is not specified

3. What is the default username for managed AD in AWS?
- Admin

4. To which security best practice does the IAM Access Analyzer apply?
- Least privilege

5. Which limitation applies to IAM group management?
- Groups cannot be nested

6. Which details must be specified when creating a custom IAM policy?
- Service
- Actions

7. Which PowerShell cmdlet is used add IAM users to an IAM group?
- Add-IAMUserToGroup

8. Which IAM resource consists of related resource permissions?
- Policy

9. Which user access options are available when creating an IAM user in the console?
- Console Access
- Programmatic access

10. You are using the console to manage IAM users. After opening an IAM user account, where should you click to enable MFA?
- Security credentials tab

11. How do Windows clients find domain controllers when joining an Active Directory domain?
- DNS

12. Which CLI command is used to show the groups a user is a member of?
- aws iam list-groups-for-user

13. To which AWS resources types can permissions boundaries be applied?
- IAM users
- IAM roles

14. Which CLI command is used to show the details of a specific IAM group?
- aws iam get-group

15. How do resource provider apps verify the validity of identity provider authentication tokens?
- They use the identity provider public key

16. You have noticed that the IAM Access Analyzer keeps reporting a finding related to an S3 bucket named BUCKET1 with public access enabled. BUCKET1 is used by a web app to serve product PDFs to website visitors. You no longer wish to see the finding related to BUCKET1. What should you do?
- Archive the finding for BUCKET1

17. Which options are available for configuring identity providers using the IAM Identity Center console?
- Managed AWS AD
- On-premises Active Directory